

Generality

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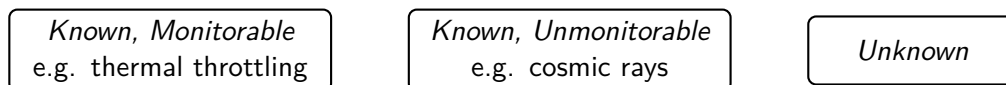
I What is Steerable Observability?

An umbrella term for different usecases:

1. *Characterizing*, to formulate a model of system behavior.
2. *Monitoring*, to flag deviations from model.
3. For observed deviations,
 - If *known*, facilitate online fixes if viable. Example: bad node found, facilitate live migration if supported. Out of scope if fix is manual (terminate, prune, restart).
 - If *unknown*, facilitate offline investigation via snapshots of system state (the *crime scene*).

I.1 Types of Anomalies

1. *Known, Monitorable* e.g. thermal throttling
2. *Known, Unmonitorable* e.g. cosmic rays
3. *Unknown*: everything else



I.2 Enabling Investigation

Investigation is fundamentally about navigating causal graphs, from symptom to cause. This creates a challenge: *what happened* can only be captured apriori. Luckily, there is a difference between the minimal and maximal definitions of *capture*:

- Capturing events to local memory
- Serializing and transporting it over a message bus, and persisting it to a database or shared storage